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MediQueue

Smart Hospital Management and Appointment System

PROJECT ABSTRACT

Project Title:	MediQueue – Smart Hospital Management and Appointment System
Domain:	Web / Mobile Application Development, Healthcare Information Systems
Project Type:	B.Tech / IT College Mini Project
Technology Stack:	HTML, CSS, JavaScript, Node.js, Express.js, MongoDB

Introduction

Healthcare institutions play a critical role in society, and the efficiency with which they manage patients, doctors, and appointments directly affects the quality of care delivered. Despite this, a large number of hospitals and clinics still depend on **manual, paper-based methods** for registration, appointment scheduling, and record-keeping. This often results in long patient waiting times, appointment conflicts, misplaced or duplicated records, and poor coordination between departments. **MediQueue** is proposed as a digital solution to these persistent problems, offering a centralized platform that brings patients, doctors, and hospital administration onto a single, organized system.

Problem Statement

Traditional hospital management relies heavily on manual registers and verbal communication, which leads to **inefficient queue handling**, difficulty in retrieving patient history, scheduling errors, and a lack of real-time visibility into appointment status for both patients and staff. There is a clear need for a system that automates these processes, reduces administrative overhead, and improves the overall patient experience.

Proposed System

MediQueue is a web/mobile-based hospital management and appointment system that digitizes core hospital workflows. The system allows patients to register and book appointments with available doctors, while doctors and administrative staff can manage schedules, view patient details, and update appointment status in real time. Key modules include **patient registration, doctor management, appointment booking and scheduling, digital patient records, real-time queue management, department-wise organization, notifications**, and an **admin dashboard** for monitoring overall hospital activity. Together, these modules replace manual coordination with a structured,

transparent digital workflow.

Technology Used

Component	Technology
Frontend (Web)	HTML, CSS, JavaScript
Mobile App	Flutter
Admin Dashboard	React.js
Backend	Node.js + Express.js
Database	MongoDB
API	RESTful APIs
Authentication	JWT + bcrypt
Push Notifications	Firebase Cloud Messaging
Version Control	Git + GitHub
Architecture	Client-Server

Expected Benefits

By digitizing hospital operations, MediQueue is expected to **reduce patient waiting time** through organized queue management, **minimize scheduling conflicts**, provide quick and reliable access to patient records, and improve communication between patients, doctors, and administrative staff. The centralized admin dashboard further enables hospital management to monitor daily activity and make informed operational decisions, resulting in a more efficient and patient-friendly healthcare experience.

Conclusion

MediQueue presents a practical and scalable solution to common hospital management challenges by replacing manual processes with a streamlined digital system. Through features such as appointment scheduling, digital records, and real-time queue management, the project demonstrates the effective application of full-stack web development concepts to solve a real-world healthcare problem, while leaving scope for future enhancements such as **telemedicine integration and advanced analytics**.

Keywords: *Hospital Management System, Appointment Scheduling, Queue Management, Digital Patient Records, Node.js, Express.js, MongoDB, Web Application*

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